

Pranav Vishwanatham

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SUMMARY

Software and AI Engineer with 3+ years of experience building complex, scalable applications and AI-powered systems. Proven expertise spanning advanced machine learning architectures, computer vision pipelines, automotive embedded software, and full-stack software development. Adept at designing resilient system architectures, managing end-to-end AI lifecycles, and leveraging cloud platforms to deploy robust, high-performance production solutions.

SKILLS

- **Programming:** Python, C/C++, HTML/CSS, JavaScript, MySQL
- **AI/ML Technologies:** Machine Learning (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn), Deep Learning (PyTorch, TensorFlow, CNN, RNN, LSTM, Transformers), Computer Vision (OpenCV, YOLO, ResNet), GenAI & NLP (LLMs, RAG, Agentic AI, MCP, MoE, LangChain, Hugging Face, Feature Engineering, Word Embeddings, spaCy, NLTK)
- **Cloud Technologies:** AWS (S3, ECR, SageMaker), Azure (AI Search, Foundry, Blob Storage)
- **Tools:** DOORS, ALM, Jira, ETAS (ASCET), MATLAB/Simulink and Stateflow, Jenkins, Splunk, Tableau, Figma, SVN, GIT, MongoDB, Twilio, FastAPI, Docker, Claude Code, GitHub Copilot
- **Automotive:** Model-Based Development (MBD), V-Model, SIL/HIL Testing, Vehicle Dynamics, Braking Systems, AUTOSAR, ASPICE, CAN Protocol, ISO26262

EXPERIENCE

AI Engineer

August 2025 – January 2026

Crevavi Inc.

Columbus, Ohio, United States

Summary: Architected and deployed advanced computer vision frameworks and data pipelines, optimizing model inference latency and improving edge-case detection accuracy for production environments. Engineered scalable MLOps orchestration and dynamic routing models, ensuring seamless integration of complex AI solutions into edge devices.

- Architected a Hybrid Router-Expert computer vision framework inspired by Mixture of Experts (MoE), using ResNet & YOLO to dynamically route difficult samples to specialized expert models improving detection accuracy on rare/edge cases by 18% while maintaining real-time performance.
- Built Python data pipelines to preprocess and reformat raw datasets into ResNet and YOLO-compatible structures, orchestrating S3 fetch/upload workflows via AWS Boto3 to support training dataset versioning across environments.
- Developed model training and validation code in PyTorch with CUDA accelerated GPU training, generating performance metrics and visualizations across precision, recall, accuracy, loss curves, and classification results to evaluate and iteratively improve model quality.
- Implemented a router+expert inference pipeline that ensembles outputs from multiple expert models via router-activated selection across 80 classes, reducing inference latency by 35% and edge device computational costs by 40%.
- Deployed MLOps orchestration components, including monitoring, model inventory practices, and validation pipelines on AWS SageMaker to support the rollout, documentation, and integration of multi-model systems into production.

Software Engineer

February 2025 – July 2025

Translation Commons

Remote, United States

Summary: Led full-stack development and technical infrastructure initiatives, streamlining data aggregation and establishing scalable open-source collaboration standards for globally distributed teams. Translated complex cross-functional stakeholder requirements into actionable technical specifications, significantly reducing onboarding overhead.

- Developed key features for the UNESCO World Atlas of Languages (WAL) website redesign, simultaneously aggregating and structuring global language datasets from authoritative sources to ensure data integrity.
- Established the technical infrastructure and GitHub organization setup, creating centralized project management and open-source collaboration standards that enabled seamless code integration for new engineering volunteers.
- Designed and led requirements engineering for a Volunteer Tracking application, translating cross-functional stakeholder needs into a technical spec that reduced manual onboarding overhead for a globally distributed team.

Associate Software Engineer

September 2021 – December 2022

Bosch Global Software Technologies

Hyderabad, India

Summary: Engineered and validated robust software for automotive embedded systems, driving cross-functional Agile delivery and

- continuous improvement in software reliability. Led critical system functions from requirements analysis through designing and executing software changes and comprehensive testing, reducing defect leakage and improving customer satisfaction.
- Developed and validated ECU software for active safety braking systems, ensuring strict adherence to functional safety governance (ISO 26262, ASPICE) and AUTOSAR architecture, enforcing technical controls and documentation standards throughout the software lifecycle.
 - Led Cooperative Regenerative Braking (CRB), Driver Brake Request (DBR) and Sensor Signal Processing (SSP) functions as Function Owner within 6 months, driving a 20% boost in customer satisfaction and 10% defect reduction.
 - Analyzed customer/system requirements and translated them into functional specifications, ensuring complete traceability through the V-Model and compliance with functional safety standards.
 - Designed and executed Model-Based software changes with full SIL/HIL testing, ensuring seamless software integration, improved reliability and preventing gaps.
 - Collaborated in cross-functional Agile teams (Scrum) to deliver production-ready software, earning “Best Newcomer” recognition for exceptional performance.

Software Engineering Intern
Robert Bosch Engineering and Business Solutions Private Limited
Summary: Developed a tool and continuous integration workflows to optimize system resource usage and enhance overall software performance. Engineered Python scripts within Jenkins CI/CD to systematically compile builds and extract metrics, utilizing predictive machine learning models and a full-stack application to significantly reduce computational overhead.

November 2020 – August 2021
Hyderabad, India

- Developed a Resource Optimizer tool to identify ESP braking system resource usage patterns and determine optimal ECU parameter configurations from software build data.
- Engineered a systematic data collection pipeline using Python and Jenkins, configuring parameterized builds to compile multiple ECU variants and extract RAM and ROM metrics from build log files.
- Integrated ECU builds with a Lab Car (HIL) testing environment to capture runtime data, consolidating RAM, ROM, and runtime metrics with parameter configurations into structured CSV and HTML reports with visualizations (Matplotlib/Plotly).
- Trained ML models on ECU build metrics stored in MongoDB to predict optimal parameter configurations, achieving a 10% reduction in resource usage; surfaced predictions via Tkinter interface enabling team to input configurations and get results.

EDUCATION

Master of Engineering in Computer Science and Applications, GPA - 3.87/4
• **Virginia Polytechnic Institute and State University (Virginia Tech)**

January 2023 – December 2024
Blacksburg, Virginia, United States

Master of Technology in Software Engineering (Integrated), GPA - 8.09/10
• **Vellore Institute of Technology (VIT University)**

July 2016 – May 2021
Vellore, India

CERTIFICATIONS

• **Microsoft Certified: Azure AI Apps and Agents Developer Associate** | [View Credential](#)

August 2026

PROJECTS

Enterprise AI Agent

June 2026

- Engineered enterprise search components for a RAG-based AI support agent utilizing Microsoft Foundry, Azure AI Search, and Blob Storage to provide students and sales teams with rapid answers on course details and pricing. Enforced strict technical controls via system instructions and negative prompt verification, ensuring securely grounded, in-scope responses. Integrated Application Insights for continuous observability and Azure Content Understanding workflows to parse structured PDF fields for downstream integration.

SmartDocs Q&A

April 2026

- Developed a high-speed RAG application using LangChain, HuggingFace embeddings, and FAISS vector search to enable interactive querying across unstructured PDF documentation. Integrated Groq’s ultra-low latency inference engine to evaluate and compare responses across open-source LLMs including LLaMA 3, Mixtral, and Gemma. Enforced grounded retrieval and auditability by surfacing source-attribution metadata (page numbers and exact document chunks).

AlertAI

September 2024

- Engineered a real-time emergency response platform utilizing FastAPI backend and an interactive Gradio dashboard to assist 911 dispatchers. Integrated Twilio, OpenAI and Azure services to enable live voice call streaming, instant speech transcription, and generate context-aware AI recommendations based on the emergency. The application leverages MongoDB for data management alongside features dynamic map integrations for active incident tracking.